Amar Prakash Pandey

Senior Data Engineer — Spark Performance & Data Platforms

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Summary

Senior data engineer with 8+ years in engineering, the last 5 owning terabyte-scale batch pipelines on **Spark and Airflow** across ad-tech on AWS EMR (Sahaj) and Market Risk on-prem (Citi). Known for root-causing expensive or failing jobs and building the tooling that keeps teams fast, from Spark log analyzers to CI/CD and test automation. **Apache Airflow contributor; former maintainer at CRI-O and DuckDuckGo.** Writes on Spark internals at amarpandey.me/blog.

Experience

Citi — Bengaluru, India

Senior Data Engineer — *June 2025 – Present*

- Took ownership of daily failing, over-provisioned Spark jobs, root-causing config issues to cut resource usage by **up to 90%**; built Spark log and Autosys JIL analyzers so the whole team diagnoses jobs faster.
- Built **var-compare** from scratch, the daily tool the team uses to verify Market Risk numbers before reports go out.
- Led the **Maven → Gradle** migration of the trading-reports mono-repo: live in 2 weeks with **zero incidents**, **∼40% faster builds**, and vulnerable dependencies removed.
- Started a testing initiative on a zero-test, ∼20K-LOC repo and took it to **60% coverage**; management adopted it as a mandate for every team (**80% minimum in CI**) and teammates joined to extend it.
- Set team engineering standards: introduced a release branching strategy, codified conventions as **Claude Code skills** to keep AI-generated code consistent, and mentored junior engineers on Spring Boot and build tools.

Sahaj Software — Pune, India

Senior Data Engineer — *March 2021 – June 2025*

- Owned batch pipelines, data quality and performance for audience data behind out-of-home (OOH) ad campaigns.
- **Designed and optimized ETL pipelines** with **Apache Spark (PySpark) on AWS EMR**, processing terabytes of data and running large-scale joins to enrich advertisement analytics.
- **Improved Spark job performance by 50%** through shuffle partition tuning and resource optimization.
- **Upgraded self-managed Airflow on AWS to a newer version, end-to-end**, with live pipelines running on it.
- Analyzed S3 usage, designed lifecycle rules and shipped them to production, **cutting monthly S3 costs by 60%**.
- Automated deployment of **Airflow DAGs**, **Spark/PySpark jobs**, and **microservices** using **GitLab CI/CD**, built with **Cursor AI** in a fraction of the estimated time with broader test and edge-case coverage. Removed manual errors across dev-to-prod promotion.

Tata Consultancy Services — Chennai, India

System Engineer — *December 2017 – February 2021*

- Worked on Cummins' **pricing platform**, which automated pricing updates for thousands of components.
- **Developed APIs** using **Java, Spring Boot, and MySQL** for real-time pricing updates.
- **Automated pricing workflows**, reducing update time from **one day to near real-time**.
- **Migrated Oracle trigger and function logic** to a service layer for better logging and maintainability.
- Automated deployment with CI/CD pipelines using Git, **Jenkins, and GitLab**.

Projects

uv-bundler — *April 2026*

- Python CLI that builds **deployment-ready Linux artifacts for Spark and Lambda pipelines from any host OS**. Resolves platform-specific wheels with uv: no Docker or Linux build server. **1.4k+ PyPI downloads.**

spot-optimizer — *January 2025*

- Python library on PyPI that picks the **most stable, cost-effective AWS Spot instances** for a given core count, memory, region, architecture and EMR version. Replaces in-house selection logic. **11k+ downloads, 14 releases, 98% test coverage, CI-published on every tag.**

geohashviz.com — *July 2024*

- Interactive tool for encoding locations into geohashes and exploring precision levels. Built to debug geohash behavior in production OOH ad pipelines.

Skills

Languages: Java, Python, Scala, SQL
Big Data: Apache Spark, PySpark, Apache Hadoop, Apache Hive, Apache Impala
Orchestration & Databases: Apache Airflow, dbt, PostgreSQL, Oracle
Cloud (AWS): S3, EMR, Athena, ECS, ECR, Lambda, RDS
DevOps: Git, GitHub, Docker, Terraform, Jenkins, GitLab CI/CD

Awards

Deliver with Pride, *Citi, for dedication and delivery on PACE work*
Recognized for Exceptional Contributions, *Sahaj Software* — *March 2024*
Star Performer of the Year, *TCS (Cummins project)* — *June 2019*
Google Summer of Code, *Teammates project, National University of Singapore* — *August 2017*